SERVER



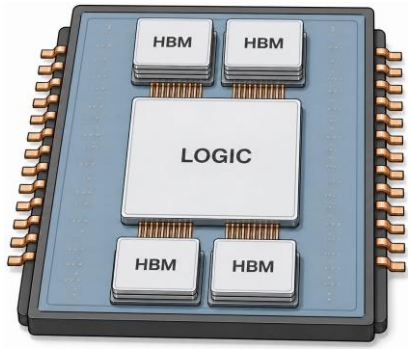
Open Silicon Photonics for AI Systems: Extending the MHS Platform Contract

Bijan Nowroozi *Head of Ecosystem Development, Lightmatter*

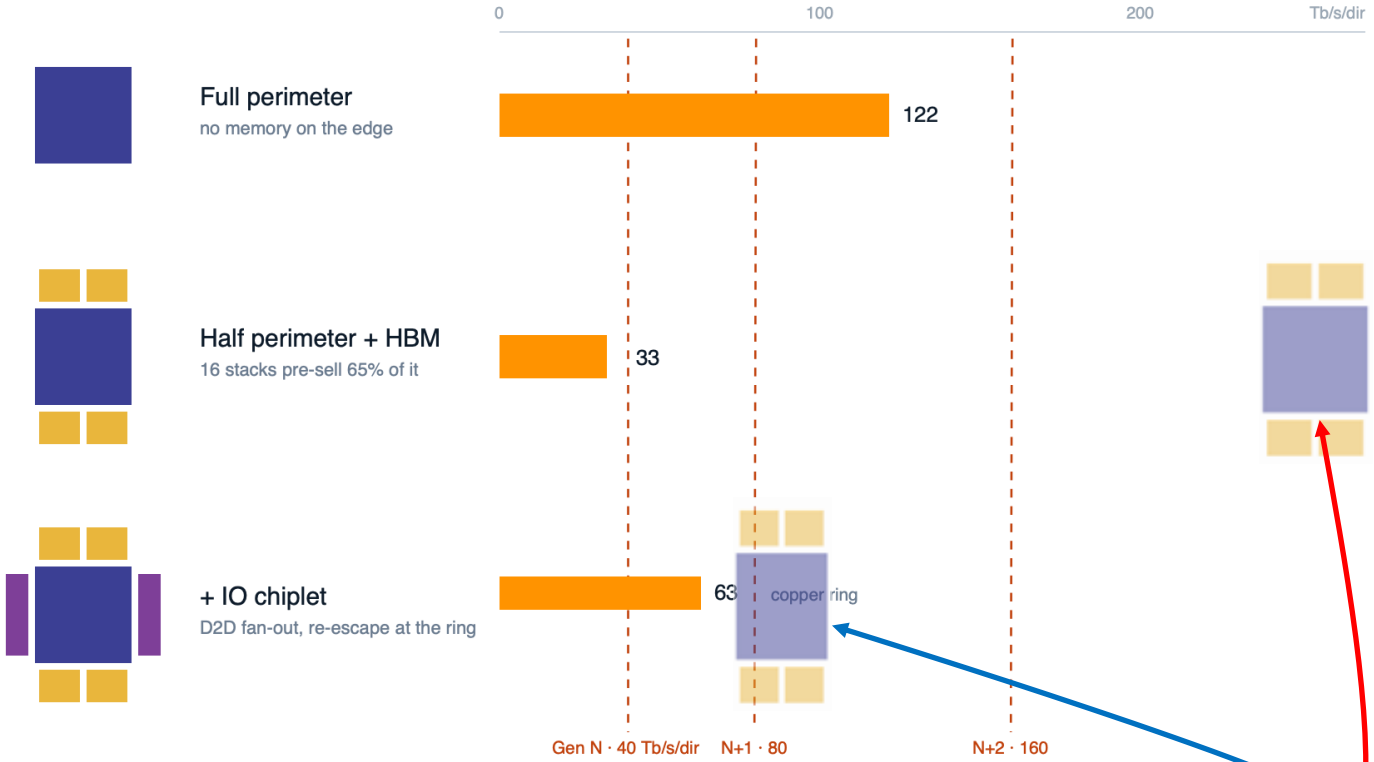
Morris Repeta *Director, Advanced Hardware Technology, Dell*



The Escape Problem Is Geometry



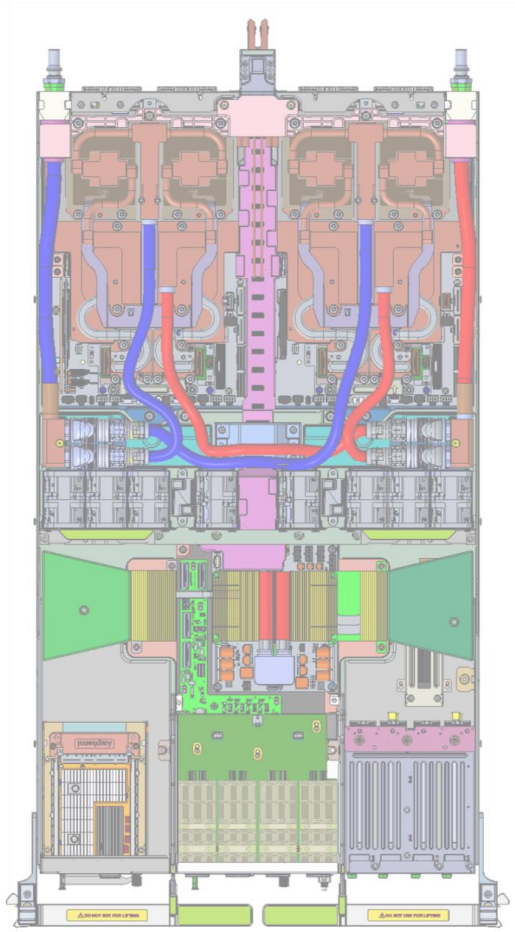
- **L vs L²:** logic grows with area, I/O escapes at the perimeter.
- **Beachfront:** HBM PHY claims the package edge first; the fabric gets the remainder.



- **Copper escape:** 26 Tb/s/dir from a reticle-class package edge at 224G.
- **CPO escape:** 93.6 Tb/s/dir from the same perimeter, ~3.6× copper.
- **Areal (APO) ceiling:** 1,191 Tb/s/dir through the die face, 46× copper.

Reticle-class package, 224G lanes. Bullets are switch-side (CPO at 0.90 usable beachfront); the chart is XPU-side, where HBM claims the edge first. APO at microbump pitch, fabric-gross. Source: Open Silicon Photonics for AI Systems whitepaper.

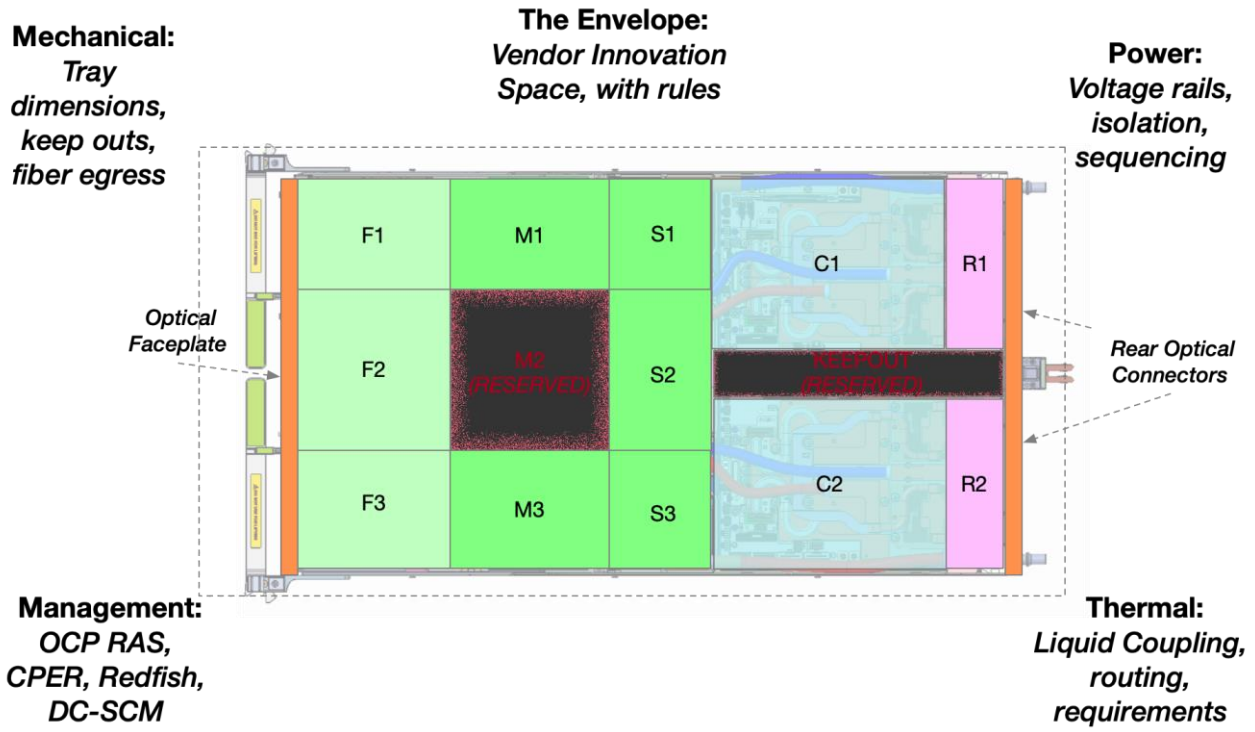
Optics Inboard: A Platform Problem



- When the optical engine leaves the pluggable cage, the tray inherits:
  - **Escape geometry —** routing and keep-outs move inboard.
  - **Faceplate —** the cage no longer defines the envelope.
  - **Internal fiber routing —** service loops, bend radius, blind-mate.
  - **Optical distribution —** laser sourcing becomes a tray resource.
  - **Power delivery —** OE rails and sequencing on the tray.
  - **Thermal —** OE and ASIC now share a solution.
  - **Management visibility —** the cage's CMIS anchor point is gone.
- The platform contract must answer all seven.

Extending DC-MHS, Not Replacing It

- **Inherited:**
  - DC-MHS modularity and tray boundaries; DC-SCM management; ORv3 power and rack.
- **Extended:**
  - Optical distribution (ELS shelf); internal fiber routing and service; OE thermal envelopes; photonic escape zones. Together they open the path to higher silicon and I/O density.
- **Unchanged:**
  - Platform-layer interoperability at the tray boundary. Engine choice stays with the vendor.



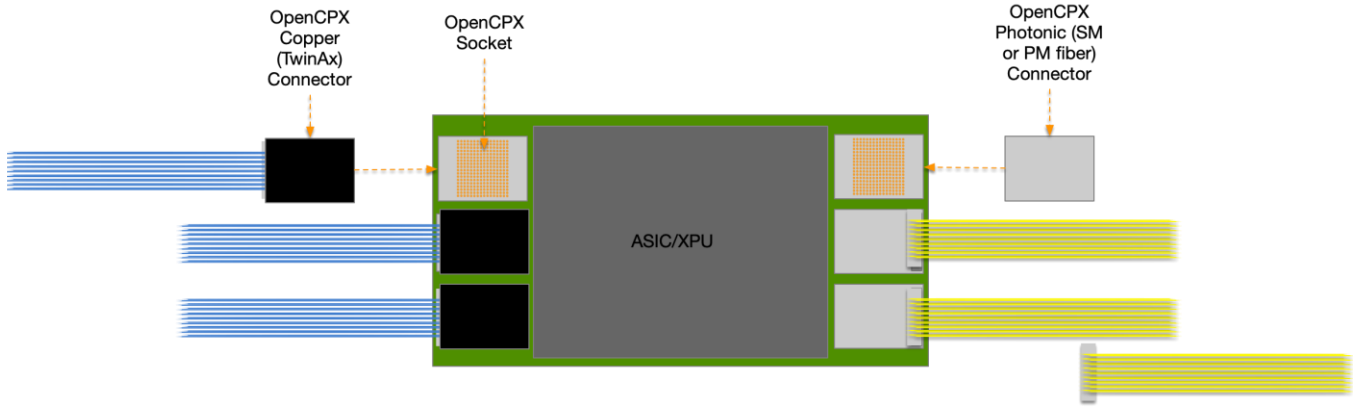
Designed for All Placement Classes

- **FPO (front-panel):** optical engine in the cage. The reference baseline and the most decoupled class.
- **NPO (near-package):** OE on the host PCB beside the package. Socketed (S-NPO) and mezzanine (M-NPO) variants move the FRU boundary.
- **CPO (co-packaged):** OE inside the XPU or switch package, 2D or 3D. No connector in the channel; the FRU is the package.
- Placement classes say where optics sit; link profiles say what runs on the fiber.
- Any engine that meets its class contract fits the tray: vendor-neutral by design.

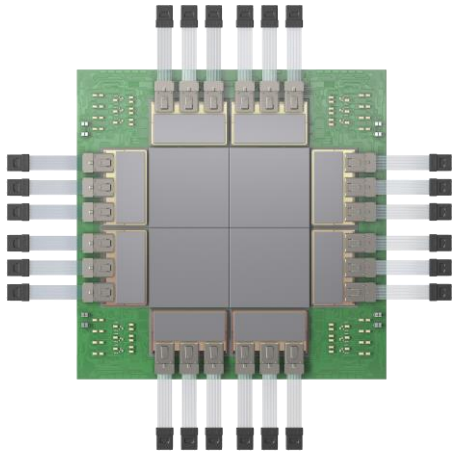
XPO



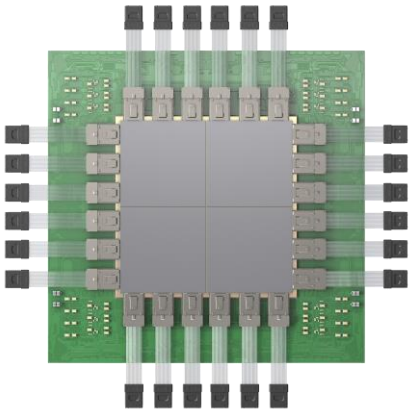
Open CPX



2D CPO



3D CPO



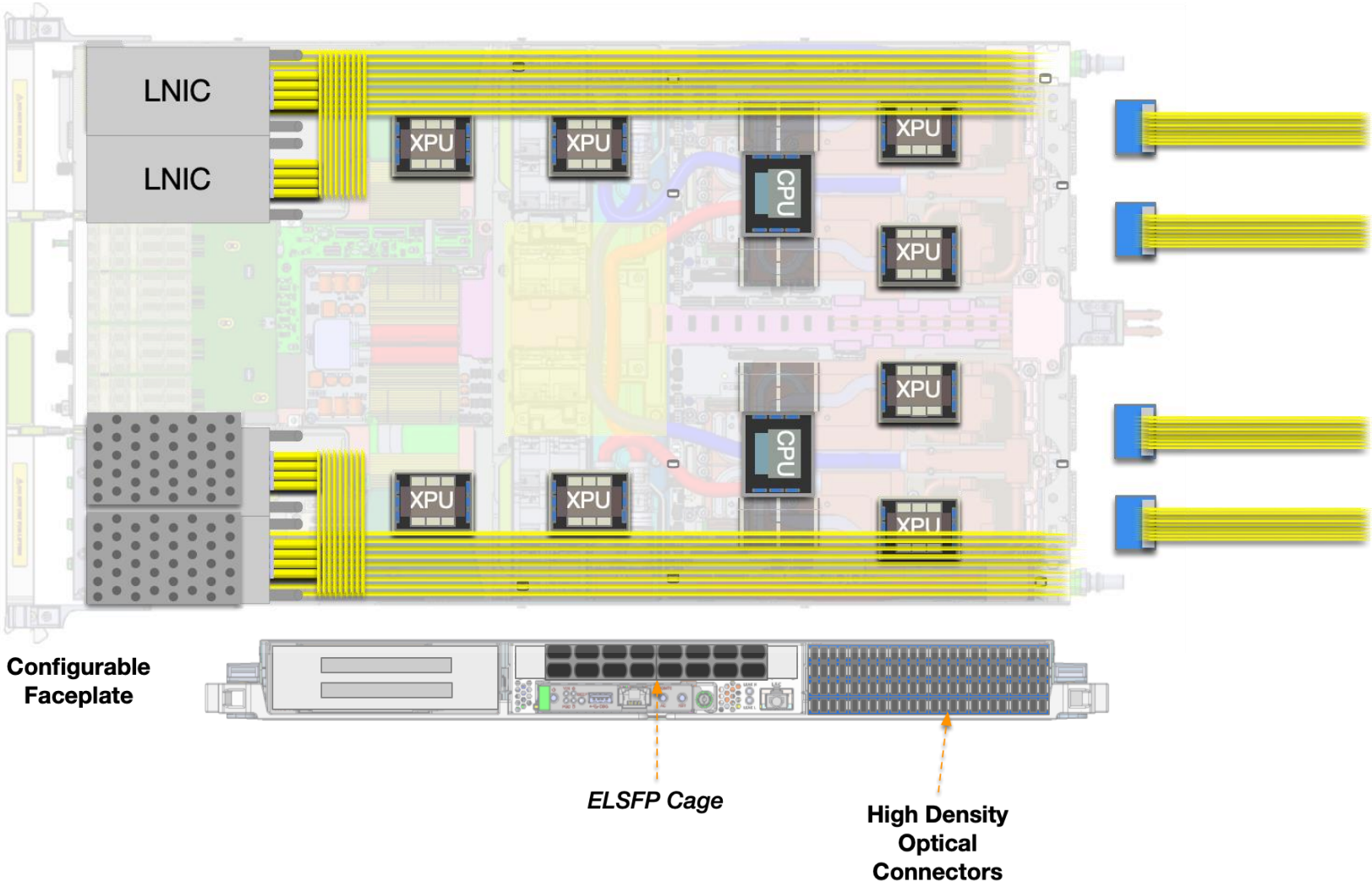
The Tray Contract: Five Dimensions

Dimension	FPO (front-panel)	NPO (near-package)	CPO (co-packaged)
Mechanical	Cage + faceplate envelope; pluggable service	OE keep-outs; fiber egress zones; socketed / mezzanine FRU variants	Package envelope; blind-mate fiber; ELS shelf; FRU = the package
Power	Cage power limits	OE rails on tray; sequencing	ELS + OE package power; sequencing
High-speed signaling	Host SerDes to cage (LR-class)	Short-reach host trace (XSR/VSR); unretimed to half-retimed	Unretimed XSR-class; no connector in the channel
Thermal	Faceplate airflow	Shared airflow, OE + ASIC	Package-level cooling; liquid-ready
Management	CMIS at the cage	CMIS + tray sensor extension	RAS telemetry via DC-SCM path

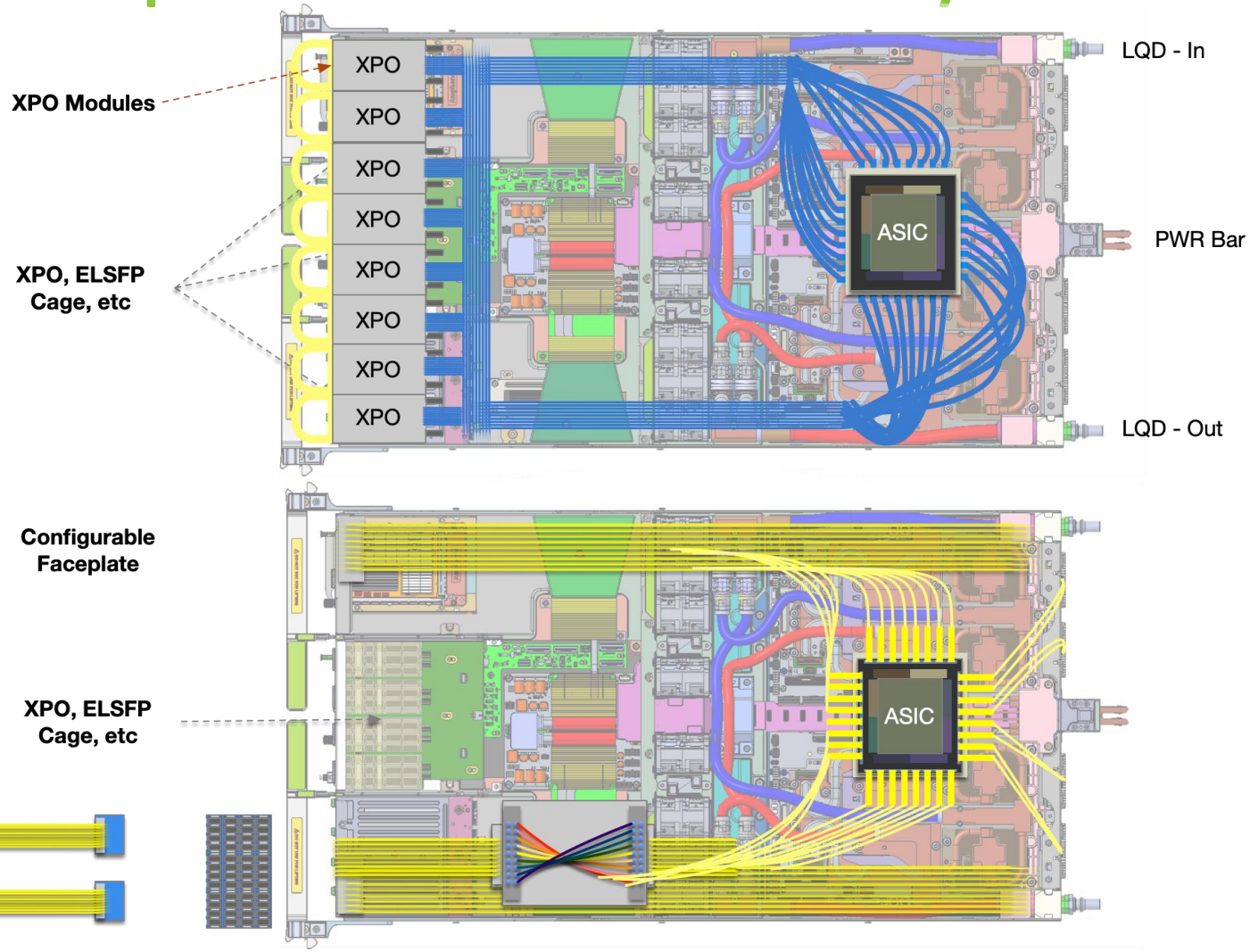
Contract dimensions per the OSP for AI Systems whitepaper; values are what the tray must specify, not implementations.

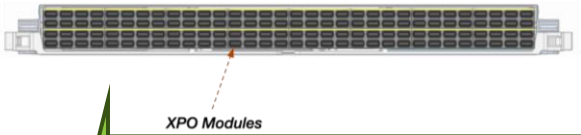
Composable Compute Tray

- High-density, multi-die XPU modules.
- Zones make the tray composable.
- Heterogeneous optical engines supported.
- Many configurations; one set of interfaces.
- >400 Tb/s of fiber routing and connectivity.

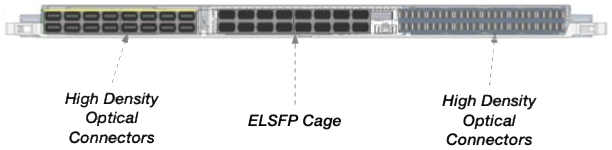


Composable Switch Trays





- Flyover copper and photonics on one tray.
- 204.8T class: 512 × 400G or 1,024 × 200G ports.
- Faceplate pluggables and CPO under one tray contract.
- 1RU density at 200T and beyond.



Management and Reliability

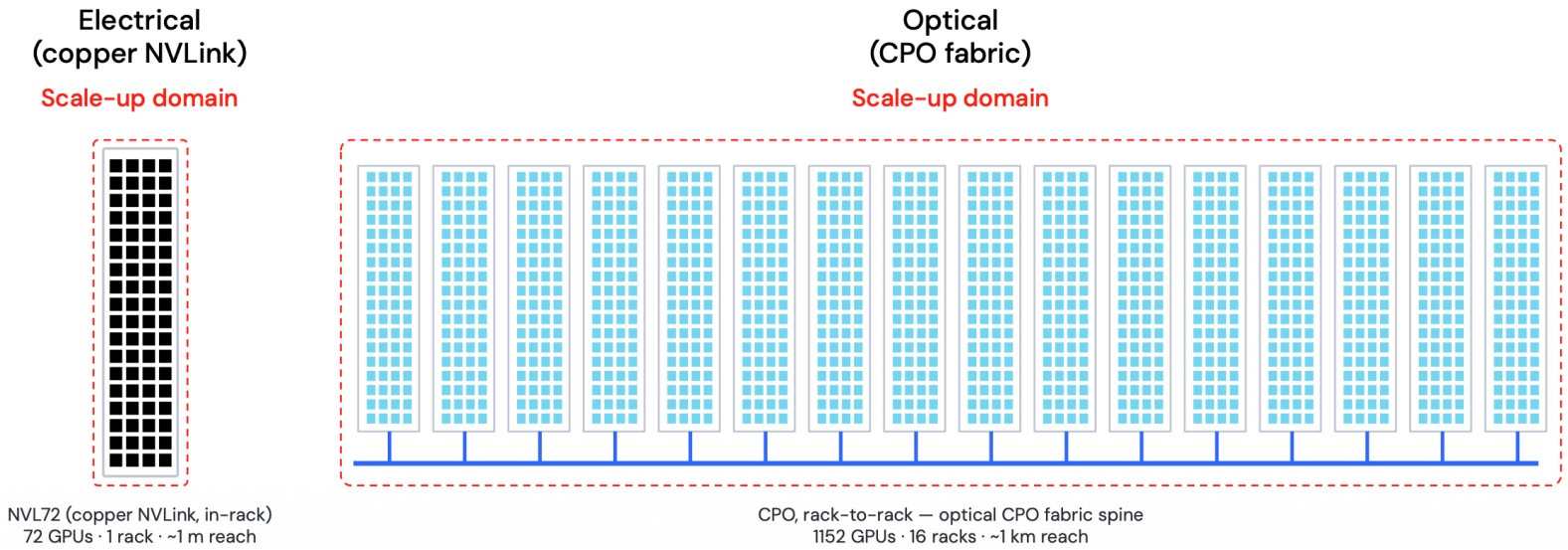
- Today the pluggable cage is the CMIS anchor. Move the OE inboard and that anchor disappears.
- The contract restores visibility:
  - OCP RAS telemetry for optical engines and lasers.
  - OIF CMIS extension path for inboard OEs.
  - DC-SCM integration — one management plane.
- Reliability is measured: first-gen CPO fleet MTBF of 8.2M hours, **~10× pluggables**; second gen past 50M operating hours (Meta, ECOC 2025 / OFC 2026).
- Serviceability: laser as FRU (ELS); fiber service loops in the mechanical spec.

Contract-Based Design Delivers

- **Utilization:** ~10× lower straggler tail on a 512-way collective; MFU target 55–65%.
- **Network power:** <5 pJ/bit for area-class photonics vs 15–25 electrical; interconnect share ~37% → ~15%.
- **Scale-up domain:** grow the flat radix first, then add sub-10 ns photonic hops instead of 40–120 ns retimer hops.
- **Deployability:** one fiber plant for Class-H and Class-V racks; no massive midplanes.
- **Future ready:** the next lane generation fits the same contract.
- **Open and serviceable:** multi-vendor trays under the OCP RAS service model.

Electrical: copper, in-rack (72 XPUs)

Optical: CPO, rack-to-rack (1,152 XPUs)

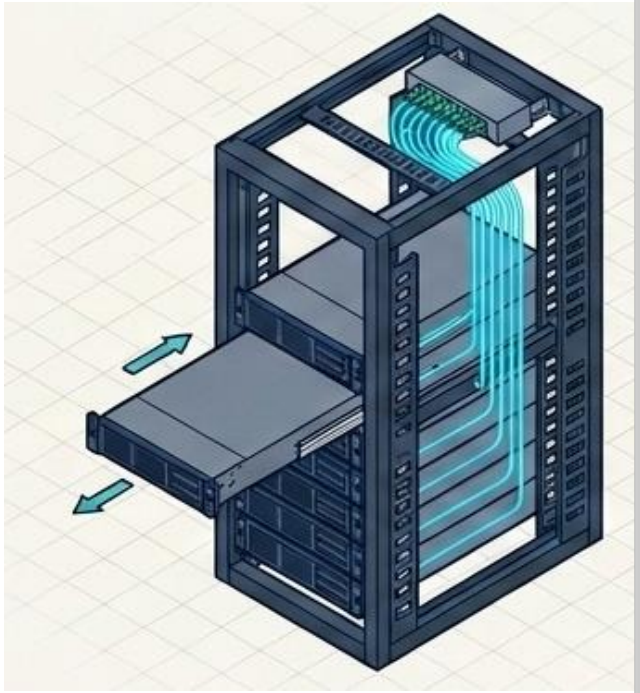


16× larger scale-up domain, from ~1,000× reach and 8× port density

Source: SemiAnalysis (2026) "Nvidia: The inference kingdom expands"

Build For Ultra Domain Sizes

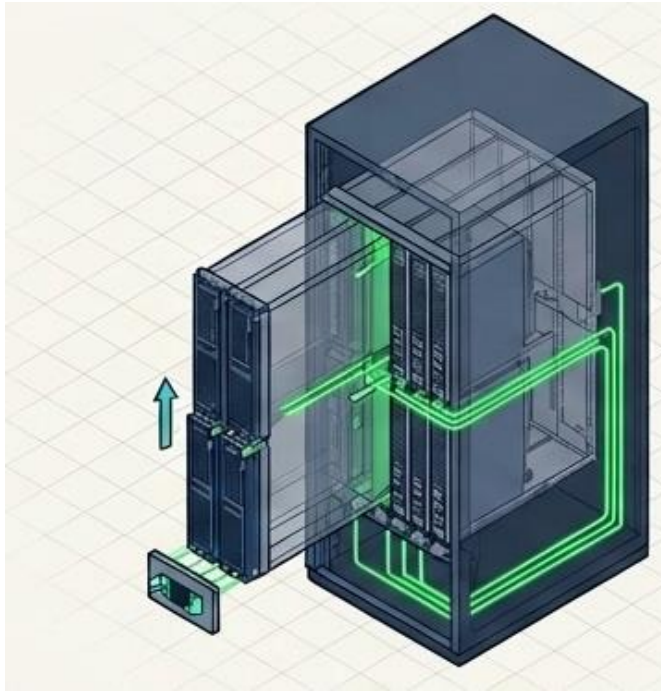
ORv3 MHS
Class-H 600kW +

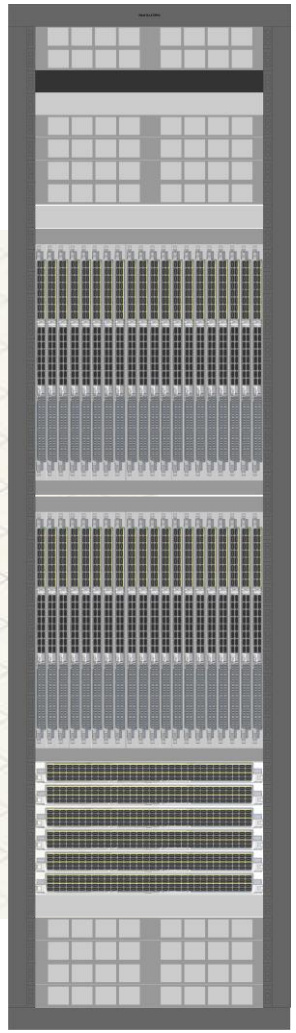




Scale-up domain: 72–144 XPUs

ORv3 Next Generation
Class-V 1MW +





Scale-up domain: 1,024–1,152+ XPUs

Core Principle: Both orientations defined with the same external interfaces.

A facility can deploy a mix of both types of racks using an identical fiber plant.

Call to Action

- **Problem to solve:** the missing layer is a contract, not a component.
- **Start here:** read and comment on the Open Silicon Photonics for AI Systems whitepaper (QR).
- **Join the workstream:** OCP MHS Interconnect-for-AI, biweekly calls. bijan@lightmatter.co
- **Help define:** link architecture · laser/RAS · mechanical integration · thermal/packaging.
- **Timeline:** whitepaper 1.0.x available now; specification drafts through 2027.
- **See it in hardware:** “High-Speed Open Silicon Photonics Reference Designs,” Wed Oct 14, 1:30 PM, Room 210AE.
- The spec is ours to co-author.

Open Discussion

